The Geometry of Fiber Bundles

Labix

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Contents

1 Fundamental Constructions

1.1 Properties of the Space of Vector Fields

Definition 1.1.1 (The Lie Bracket)
Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Define the Lie bracket of X and Y to be the vector field $[X, Y] \in \mathfrak{X}(M)$ given by

$$[X, Y]_p(f) = X_p(Y_p(f)) - Y_p(X_p(f))$$

for all $f \in \mathcal{C}^\infty(M)$ for each $p \in M$.

Lemma 1.1.2
Let M be a smooth manifold. Then $(\mathfrak{X}(M), [-, -])$ is a Lie algebra.

Proposition 1.1.3
Let M be a smooth manifold. Let G be a Lie group acting smoothly on M. Then the map $\text{Lie}(G) \to \mathfrak{X}(M)$ given by

$$\xi \mapsto \xi_M$$

is a Lie algebra homomorphism.

1.2 The Normal Bundle

Definition 1.2.1 (The Normal Bundle)
Let M, N be smooth manifolds. Let $\iota : N \hookrightarrow M$ be a smooth embedding. Define the normal bundle of N in M to be the quotient bundle

$$N_{M/N} = \frac{TM}{TN}$$

Example 1.2.2
Let $\iota : S^n \to \mathbb{R}^{n+1}$ be the standard inclusion of the unit sphere. Then the following are true.

- $N_{\mathbb{R}^{n+1}/S^n} \cong \varepsilon$ is a trivial bundle of rank 1.
- $w_i(TS^n) = 1$ for all i.

Proof
Notice that $N_{\mathbb{R}^{n+1}/S^n}$ is a bundle of rank equal to 1. It is the trivial bundle if and only if it admits a nowhere vanishing section. Define a section $s : S^n \to N_{\mathbb{R}^{n+1}/S^n}$ by $s(x) = (x, x)$ where $x \in N_{\mathbb{R}^{n+1}/S^n}$ is a vector in the direction x. Then clearly it is non-vanishing.

Now we have $TS^n \oplus N_{\mathbb{R}^{n+1}/S^n} = T\mathbb{R}^{n+1}$. Applying the total Stiefel-Whitney class gives $w(TS^n) \cdot 1 = w(TS^n)w(N_{\mathbb{R}^{n+1}/S^n}) = w(T\mathbb{R}^{n+1}) = 1$. Hence $w(TS^n) = 1$. Since $w_0(TS^n) = 1$, the grading in the cohomology ring implies that $w_i(TS^n) = 0$ for all i. ∎

2 Smooth Fiber Bundles

2.1 Basic Structures

Definition 2.1.1 (Smooth Fiber Bundles)
Let E, M, F be smooth manifolds. Let $p : E \to M$ be a fiber bundle with fiber F and local trivialization charts (U_k, ϕ_k) for $k \in I$. We say that p is a smooth fiber bundle if the following are true.

- p is smooth.
- ϕ_k is a diffeomorphism for all $k \in I$.

Lemma 2.1.2
Let $p : E \to M$ be a smooth fiber bundle with fiber F. Let $p, q \in M$. If M is connected, then E_p is diffeomorphic to E_q.

Definition 2.1.3 (Smooth Sections)

2.2 The Jet Bundle

Definition 2.2.1 (Jet Equivalence)
Let M, N be smooth manifolds. Let $f, g : M \to N$ be smooth maps. Let $p \in M$ such that $f(p) = g(p)$. Let $k \in \mathbb{N}$. We say that f and g are k-jet equivalent at p if there exists charts $(U, \phi = (x^1, \dots, x^n))$ of M and $(V, \psi = (y^1, \dots, y^m))$ of N such that $U \subseteq f^{-1}(V) \cap g^{-1}(V)$, we have

$$\frac{\partial}{\partial x_{i_j}} \cdots \frac{\partial}{\partial x_{i_1}}\bigg|_{\phi(p)} (y^t \circ f \circ \phi^{-1}) = \frac{\partial}{\partial x_{i_j}} \cdots \frac{\partial}{\partial x_{i_1}}\bigg|_{\phi(p)} (y^t \circ g \circ \phi^{-1})$$

for all $1 \leq t \leq m$ and all $0 \leq j \leq k$ and all $1 \leq i_1, \dots, i_j \leq n$. In this case we denote $f \sim_p^k g$.

Definition 2.2.2 (The Set of Jets)
Let M, N be smooth manifolds. Let $p \in M$. Define the set of k-jets from M to N by

$$J_p^k(M, N) = \frac{\{f : M \to N \mid f \text{ is a smooth map }\}}{\sim_p^k}$$

Definition 2.2.3 (The Jet Bundle)
Let M, N be smooth manifolds. Define the kth jet bundle from M to N to be the set

$$J^k(M, N) = \coprod_{p \in M} J_p^k(M, N)$$

together with the following data. For $(U, \phi = (x^1, \dots, x^n))$ a chart of M and $(V, \psi = (y^1, \dots, y^m))$ a chart of N, define

$$W(U, V) = \{[f] \in J_p^k(M, N) \mid p \in U, f(p) \in V\}$$

and define the map $W(U, V) \to \phi(U) \times \mathbb{R}^n$ by

$$[f] \mapsto \left(\phi(p), \left(\frac{\partial}{\partial x_{i_j}} \cdots \frac{\partial}{\partial x_{i_1}}\bigg|_{\phi(p)} (y^t \circ f \circ \phi^{-1})\right)_{1 \leq t \leq \dim(N), 0 \leq j \leq k, 1 \leq i_1, \dots, i_j \leq \dim(M)}\right)$$

where $n = \dim(N) \sum_{r=0}^k \binom{\dim(M) + r - 1}{r}$.

Lemma 2.2.4

Let M, N be smooth manifolds. Then $J^k(M, N)$ is a smooth vector bundle over M of dimension

$$\dim(J^k(M, N)) = \dim(M) + \dim(N) \sum_{r=0}^{k} \binom{\dim(M) + r - 1}{r}$$

Example 2.2.5
Let $k = 1$. Then we have

$$\dim(J^1(M, N)) = \dim(M) + \dim(N)\,(1 + \dim(M))$$

A local chart of $J^1(M, N)$ is given as follows. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart of M and $(V, \psi = (y^1, \dots, y^m))$ a chart of N. Then

$$\left(W(U, V), \left(\phi = (x^1, \dots, x^n), \psi = (y^1, \dots, y^m), \left(\frac{\partial \left(y^t \circ - \circ \phi^{-1}\right)}{\partial x_i} \bigg|_{\phi(p)} \right)_{1 \leq t \leq \dim(N), 1 \leq i \leq \dim(M)} \right) \right)$$

is a chart of $J^1(M, N)$.

Definition 2.2.6 (Jet Prolongation)
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Define the kth jet prolongation of f to be the section $j^k f \in \mathcal{C}^\infty(J^1(M, N))$ given by

$$p \mapsto [f] \in J^k_p(M, N)$$

Definition 2.2.7 (The Jet Bundle of a Fiber Bundle)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth fiber bundle. Define the kth jet bundle of E to be the subbundle

$$J^k(E) = \coprod_{p \in M} J^k_p(E) \subseteq J^k(M, E)$$

where $J^k_p(E) = \{[f] \in J^k_p(M, E) \mid f \in \mathcal{C}^\infty(E) \text{ is a smooth section } \}$.

3 Smooth Principle Bundles

3.1 Basic Structures

Proposition 3.1.1
Let M be a smooth manifold. Let G be a Lie group with a free and smooth action on M. Then the following are true.

- For each $x \in M/G$, there exists a neighbourhood U of x and a map $s : U \to M$ such that $\pi \circ s = \text{id}_U$, where $\pi : M \to M/G$ is the projection map.
- The projection map $M \to M/G$ is a principal G-bundle.

Example 3.1.2
Let G be a Lie group. Let $H \leq G$ be a Lie subgroup. Then G acts on H by left multiplication. The above prp shows that G/H is a Lie group and $G \to G/H$ is a principal G-bundle since the orbit space of G acting on H is just G/H.

Theorem 3.1.3 (Poincaré-Hopf Theorem)
Let M be a smooth manifold. Then we have

$$e(TM)([M]) = \chi(M)$$

3.2 Connections on a Principal Bundle

Definition 3.2.1 (The Vertical Bundle)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Define the vertical bundle of P to be the smooth subbundle

$$VP = \coprod_{p \in P} \ker(d\pi_p)$$

Lemma 3.2.2
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let $p \in P$. Then there is an isomorphism

$$\iota_p : \text{Lie}(G) \stackrel{\cong}{\to} (VP)_p$$

given by $g \mapsto \left.\frac{de^{tg} \cdot p}{dt}\right|_{t=0}$.

Definition 3.2.3 (Horizontal Distributions)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. A horizontal distribution on P is a subbundle $HP \subseteq TP$ such that the following are true.

- $TP = VP \oplus HP$.
- $(HP)_{g \cdot p} = (L_g)_*(HP)_p$ for all $p \in P$ and $g \in G$.

Generalizes connections on vector bundles in the following way: For $p : E \to M$ a smooth vector bundle, a connection on $\text{Fr}(E)$ as a principal $\text{GL}(n, F)$ bundle is the same data as a connection on $p : E \to M$ as a vector bundle.

Definition 3.2.4 (Connection Forms on Principal Bundles)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. A connection 1-form on P is a smooth section $\omega \in \mathcal{C}^\infty(T^*P \otimes \text{Lie}(G))$ such that the following are true.

- We have $\omega_p(\iota_p) = \text{id}_{\text{Lie}(G)}$ for all $p \in P$.

- We have
$$\omega_{g\cdot p}((dL_g)_p(v)) = \text{Ad}(g)(\omega_p(v))$$
for all $g \in G$, $p \in P$ and $v \in T_pP$.

Proposition 3.2.5
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Then the following are true.
- Suppose that HP is a horizontal distribution on P. Define a 1-form ω pointwise by $\omega_p(X) = \iota_p^{-1}(X^V)$ where X^V is the vertical component of X in the decomposition $T_pP = (VP)_p \oplus (HP)_p$. Then ω is a connection 1-form.
- Suppose that ω is a connection 1-form on P. Then $\coprod_{p\in P} \ker(\omega_p)$ is a horizontal distribution on P.

Definition 3.2.6 (The Curvature Form of a Connection)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let ω be a connection 1-form. Define the curvature form of ω to be the 2-form $\Omega \in \mathcal{C}^\infty(\Lambda^2 T^*P \otimes \text{Lie}(G))$ given by
$$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega]$$

This definition is motivated by the local description of the curvature form for vector bundles.

Using tensor algebra, we see that $T^*P \otimes T^*P \otimes \mathfrak{g} \cong TP \times TP \to (P \times \mathfrak{g})$. Since $\Lambda^2 T^*P$ is the subbundle of $T^*P \otimes T^*P$ consisting of alternating forms, we may also think of $\Omega \in \mathcal{C}^\infty(\Lambda^2 T^*P \otimes \mathfrak{g})$ as a map $TP \times TP \to P \times \mathfrak{g}$ satisfying $\Omega(X, X) = 0$. Also, for each $p \in P$, $\Omega_p : T_pP \times T_pP \to \mathfrak{g}$ is then an alternating bilinear map.

Definition 3.2.7 (Components of the Curvature Form)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let ω be a connection 1-form. Let Ω be the curvature form. Let $e_1, \dots, e_n$ be a basis of $\mathfrak{g}$. Define the components of Ω to be the 2-forms $\Omega^i \in \Omega^2(P)$ such that
$$\Omega = \sum_{i=1}^n \Omega^i \otimes e_i$$

3.3 The Chern-Weil Homomorphism

Let V be a finite dimensional vector space over a field k. Recall from Rings and Modules that there is an isomorphism of graded F-algebras $S(V) \cong k[x_1, \dots, x_n]$.

Definition 3.3.1 (Ad-Invariant Polynomials)
Let G be a Lie group. Let $f \in S(\text{Lie}(G)^*)$. We say that f is ad-invariant if $f(\text{Ad}_g(X)) = X$ for all $X \in \text{Lie}(G)$.

Definition 3.3.2 (The Algebra of Ad-Invariant Polynomials)
Let G be a Lie group. Define $\mathbb{R}$-algebra of ad-invariant polynomials of G by
$$I(G) = \{ f \in S(\text{Lie}(G)^*) \mid f \text{ is ad-invariant } \}$$

Let $f \in S(\text{Lie}(G)^*)$ be a polynomial. Let Ω be a curvature 2-form. For each $p \in P$, we know that Ω_p is an alternating bilinear map from T_pP to $\text{Lie}(G)$.

Definition 3.3.3 (Evaluating Polynomials on the Curvature)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let ω be a connection 1-form. Let $f \in I(G)$ be an ad-invariant polynomial of the form $f = \sum_{i_1,\dots,i_n} \alpha_{i_1,\dots,i_n} x_1^{i_1} \cdots x_n^{i_n}$. Let Ω be the curvature 2-form of ω. Define the evaluation of f on Ω to be the $2k$-form $f(\Omega) \in \Omega^{2k}(M)$ given by

$$f(\Omega) = \sum_{i_1,\dots,i_n} \alpha_{i_1,\dots,i_n} (\Omega^1)^{\wedge i_1} \wedge \cdots \wedge (\Omega^n)^{i_n}$$

Lemma 3.3.4
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let $f \in I(G)$ be an ad-invariant polynomial. Let ω be a connection 1-form. Let Ω be the curvature form. Then we have

$$df(\Omega) = 0$$

Lemma 3.3.5
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let $f \in I(G)$ be an ad-invariant polynomial. Let ω, ω' be connection 1-forms. Let Ω and Ω' be the curvature forms respectively. Then we have

$$[f(\Omega)] = [f(\Omega')] \in H^{2k}_{\text{dR}}(M)$$

Definition 3.3.6 (The Chern-Weil Homomorphism)
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let Ω be the curvature form of any connection. Define the Chern-Weil homomorphism to be the map

$$P(\Omega) : S(\text{Lie}(G)^*) \to H^*_{\text{dR}}(M)$$

given by $f \mapsto [f(\Omega)]$.

Lemma 3.3.7
Let M be a smooth manifold. Let G be a Lie group. Let $\pi : P \to M$ be a smooth principal bundle over G. Let Ω be the curvature form of any connection. Then $P(\Omega)$ is an algebra homomorphism.

Proposition 3.3.8
Let M be a smooth manifold. Let $\pi : P \to M$ be a smooth vector bundle of dimension n over $\mathbb{C}$. Let Ω be the curvature form of any connection. Suppose that $f_0, \dots, f_n \in S(\mathfrak{gl}(n, \mathbb{C})^*)$ are such that $c_X(\lambda) = \sum_{i=0}^n f_i(X)\lambda^{n-i}$ Then the following are true.

- $f_0, \dots, f_n$ generate $S(\mathfrak{gl}(n, \mathbb{C})^*)$ as a $\mathbb{C}$-algebra.
- $[f_1(\Omega)], \dots, [f_n(\Omega)]$ are the Chern classes of P.

Proposition 3.3.9
Let M be a smooth manifold. Let $\pi : P \to M$ be a smooth vector bundle of dimension n over $\mathbb{R}$. Let Ω be the curvature form of any connection. Suppose that $f_0, \dots, f_n \in S(\mathfrak{gl}(n, \mathbb{R})^*)$ are such that $c_X(\lambda) = \sum_{i=0}^n f_i(X)\lambda^{n-i}$ Then the following are true.

- $f_0, \dots, f_n$ generate $S(\mathfrak{gl}(n, \mathbb{R})^*)$ as a $\mathbb{R}$-algebra.
- $[f_1(\Omega)], \dots, [f_n(\Omega)]$ are the Pontrjagin classes of P.